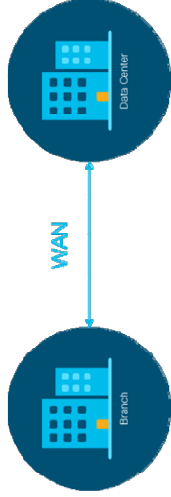


Secure Software Defined – Wide Area Network

G Srinivasu

What is SD-WAN



The traditional WAN (Wide-Area Network) function was to connect users at the branch or campus to applications hosted on servers in the data center. Typically, dedicated MPLS circuits were used to help ensure security and reliable connectivity. This doesn't work in a cloud-centric world.

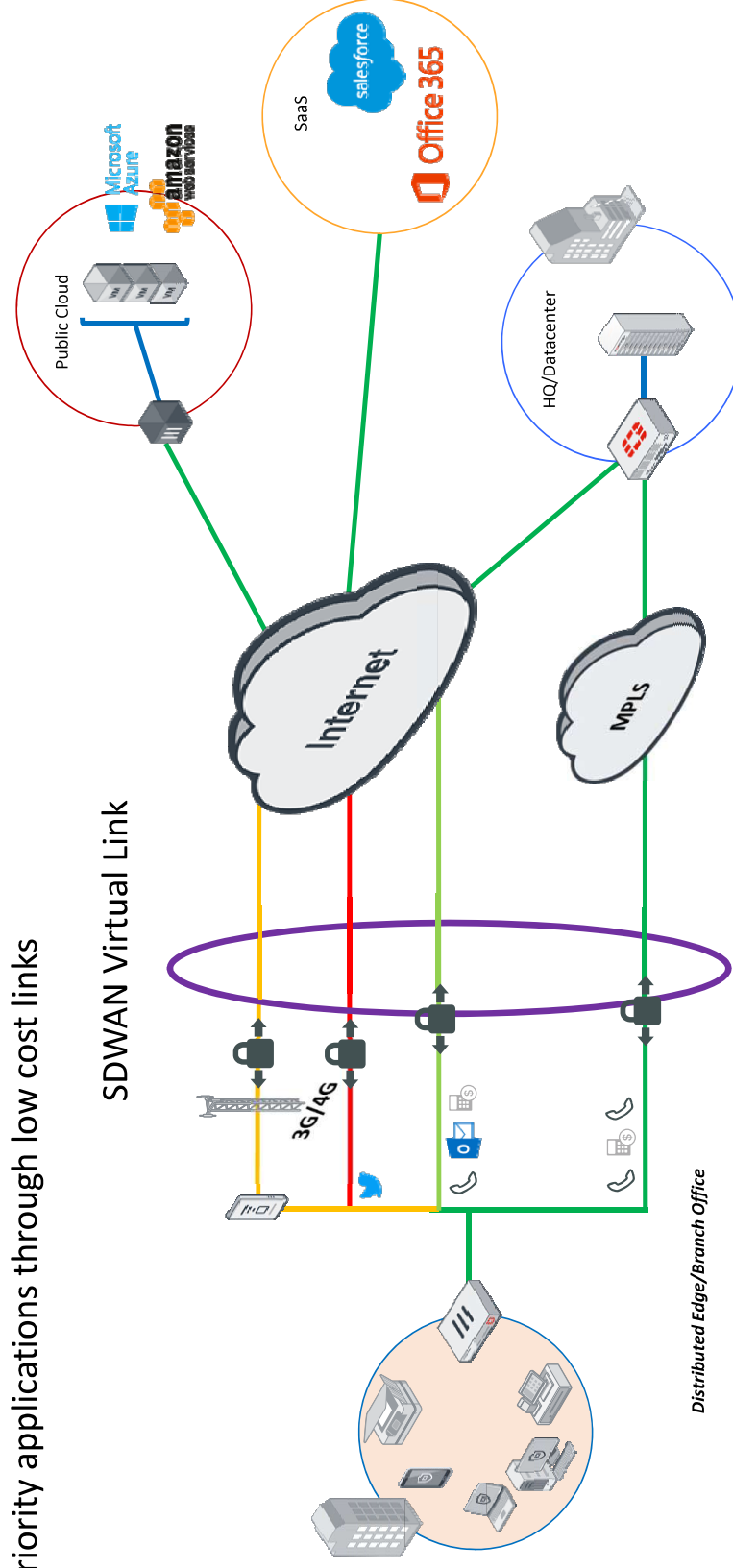
WANs designed for a different era are not ready for the unprecedented explosion of WAN traffic that cloud adoption brings. That traffic causes management complexity, application-performance unpredictability, and data vulnerability.

Further opening the enterprise to the Internet and the cloud exposes major threat and compliance issues.

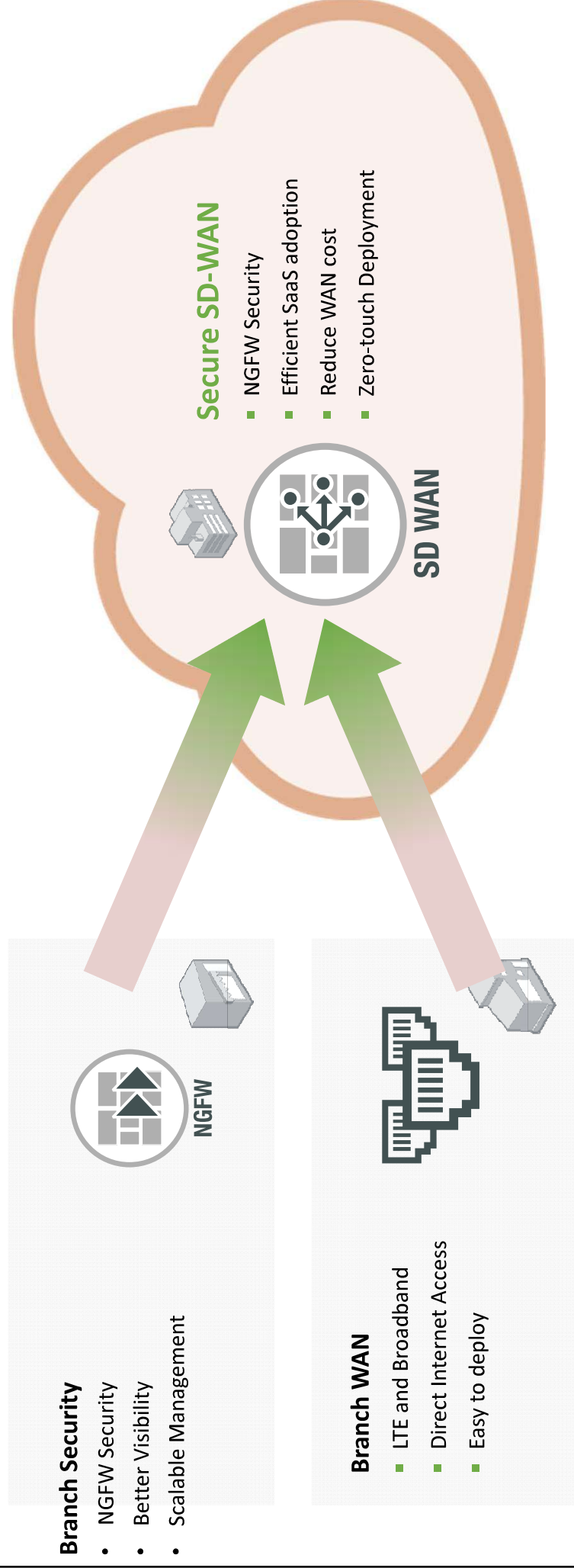
SD-WAN is a software-defined approach to managing the WAN.

SDWAN – Link Quality Measurement

- Dynamic Routing based on Link quality measurements
- Maintain High availability of Business critical applications
- Best effort for low priority applications through low cost links



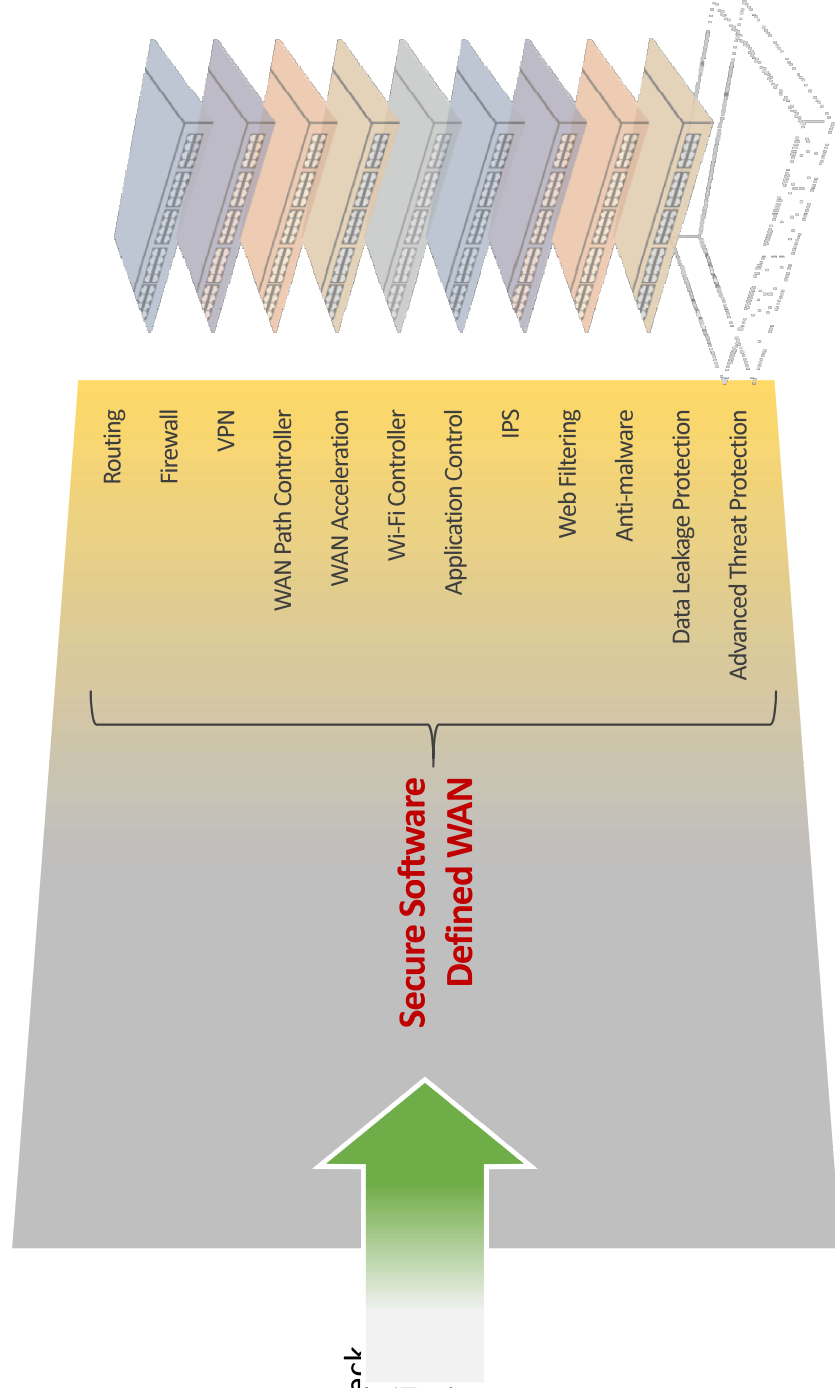
Increasing Need for secure SDWAN



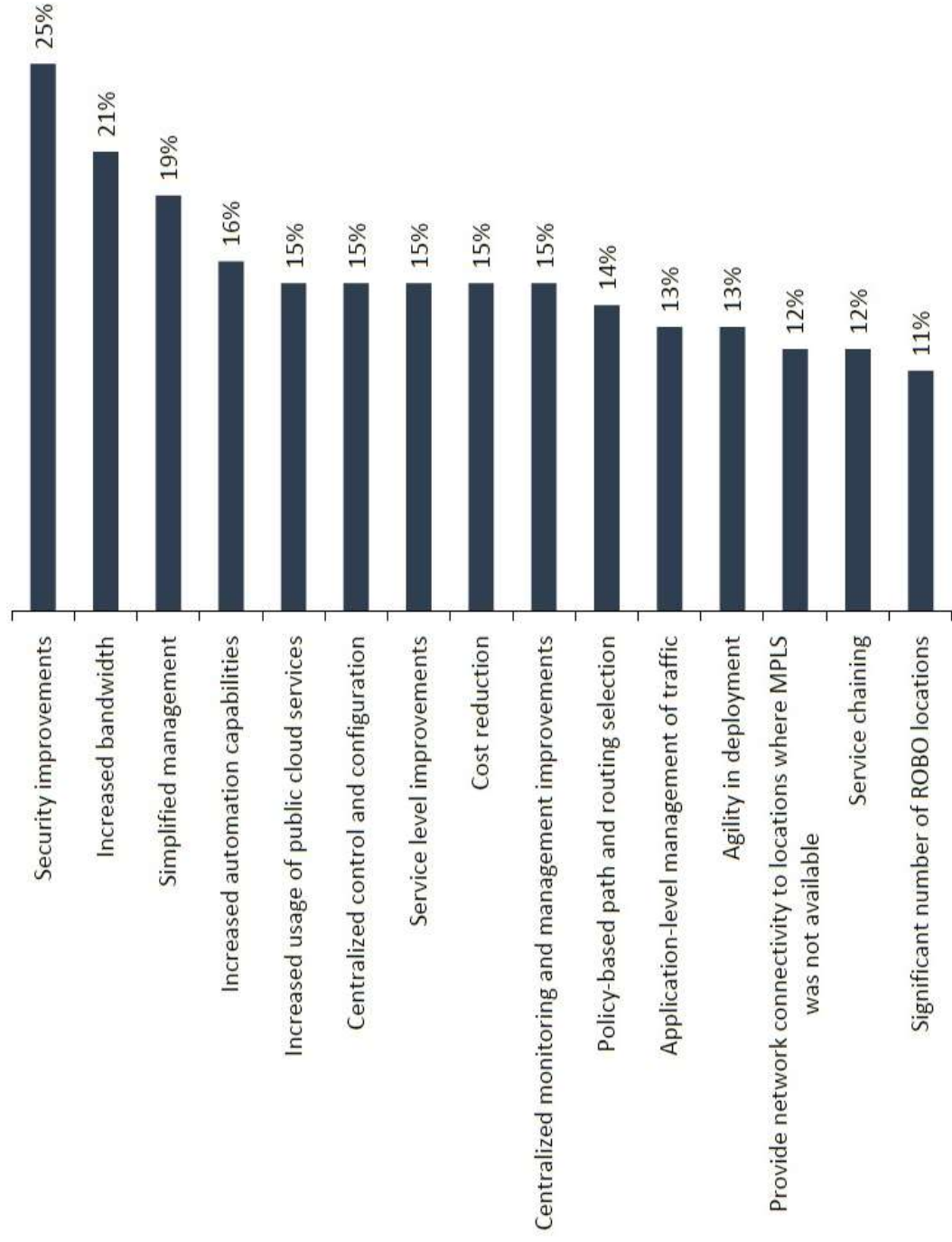
Secure SD-WAN features

SDWAN Features

- Routing and VPN
- Link Load Balancing
- Link Quality Status Check
- Packet Loss, Latency, J
- SaaS Based Path Selec

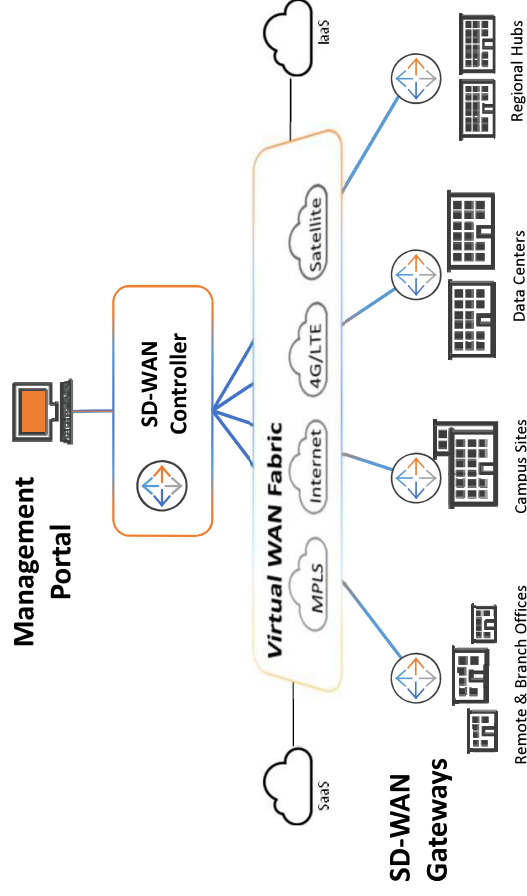


What were the most compelling reasons for your organization to adopt or consider an SD-WAN?



D-WAN: How does it work?

1. WAN Transport Virtualization
2. Centralized Management
3. App-centric & Policy-based Orchestration



Management Portal

- Single-pane of glass for configuration network management: cloud-hosted and on-premises
- Zero-touch provisioning
- Portal where user defines global policies
- Aggregated visibility on application performance

SD-WAN Controller

- Establishes and manages Secure Virtual Overlay to sites.
- Interprets the Global Policy according to the knowledge collected from the network
- Defines service chains to enable policy goals
- Manages distribution of interpreted Policy to individual elements

Virtual WAN Fabric

- Secure Overlay Fabric
- Transport Independence
- Network Segmentation

SD-WAN Gateway

- Local Policy Enforcement
- WAN Path Selection, Network QoS, Firewall, Service Chaining
- Telemetry feedback loop to/from SD-WAN Controller

SD-WAN: How does it work?

SDWAN edge includes sophisticated path conditioning techniques that can reconstruct packets lost in transit, avoiding re-transmission. SDWAN edge overcomes performance issues of the underlying transport rather than simply routing traffic around the problem. This ultimately provides better resource utilization, higher application performance, improved productivity, a better user experience and lower bandwidth costs.

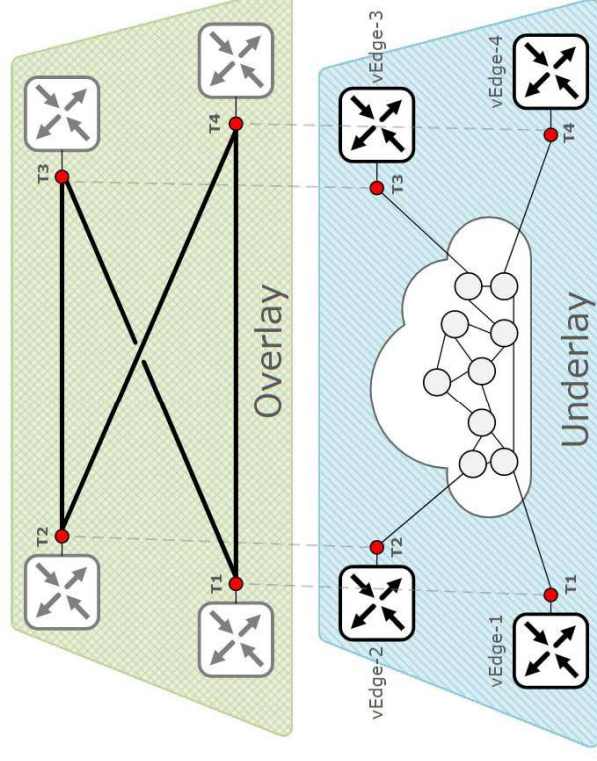
Traffic is routed using all available bandwidth on multiple links. But should a link fail, the remaining paths keep the application up and your business running.

Data deduplication and compression minimize transmission of repetitive data. This frees up bandwidth for other application traffic which saves even more money

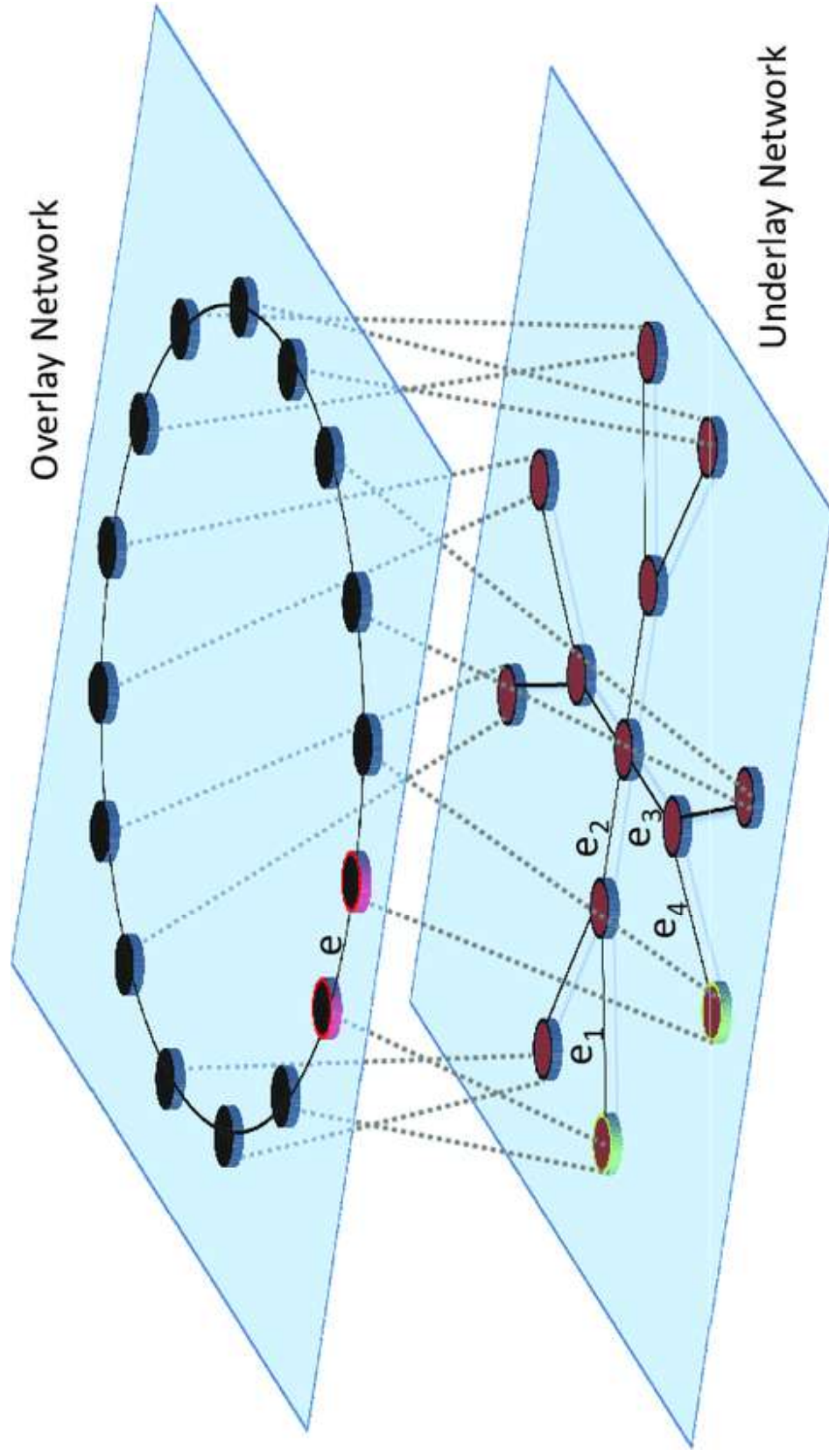
Creating an SD-WAN using virtual overlays allows micro-segmentation to extend beyond the data center. For example, a virtual WAN overlay can be created to transport a financial application with specific QoS and security requirements while isolating and handling guest Wi-Fi traffic across a different virtual overlay.

Overlay vs underlay in summary

An Underlay network is the physical network responsible for the delivery of packets like DWDM, L2, L3, MPLS, or internet, etc. while an overlay is a logical network that uses network virtualization to built connectivity on top of physical infrastructure using tunneling encapsulations such as VXLAN, GRE, IPsec.



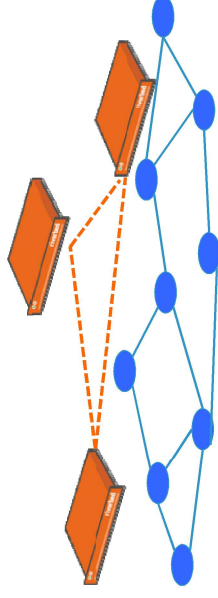
Example of Overlay on Underlay



Applications - Not Just Packets



Simple to Use

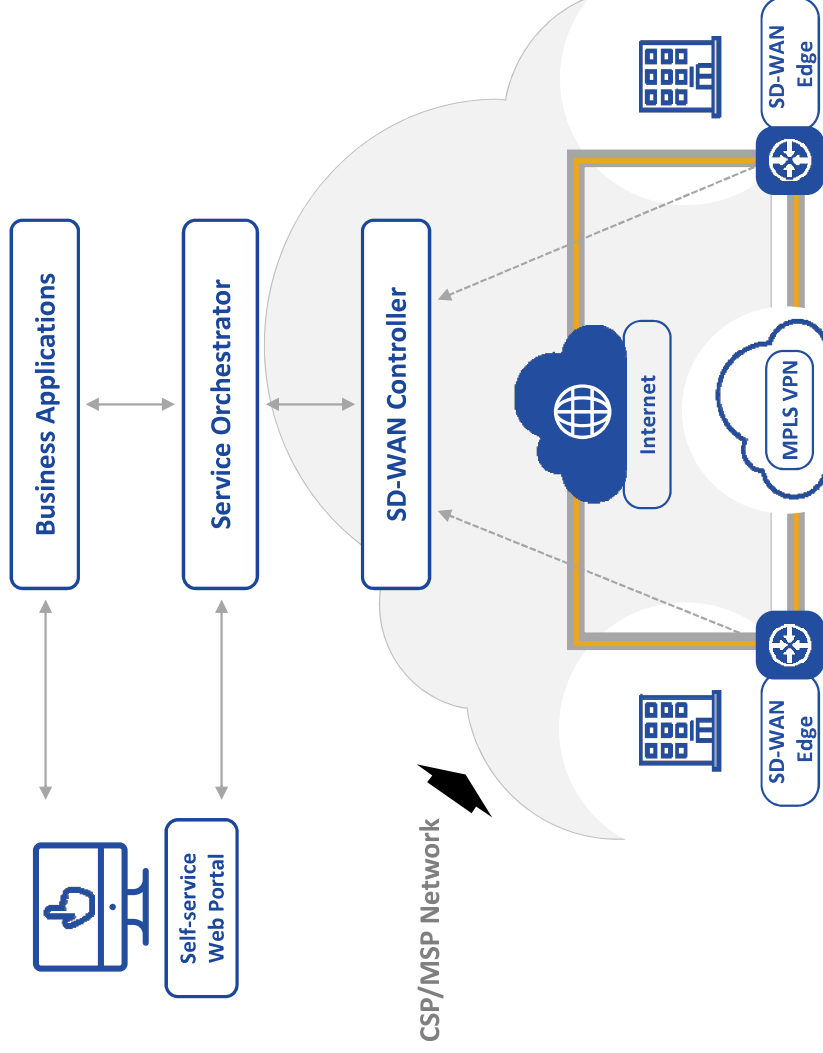


Integrated Visibility



SD-WAN Service Use Case

Hybrid WAN: SD-WAN Service over Internet and MPLS WANs



- Encrypted SD-WAN tunnel over the Internet
- Can often increase site-to-site bandwidth at no additional cost
- Increased network availability and resiliency
- Internet and MPLS VPNs can be provided by different service providers

SASE Promises

Secure Access Service Edge

- ♦ Secure Access to services for all users from anywhere
- ♦ Data Loss Prevention (DLP)
- ♦ Cloud Access Service Broker (CASB)
- ♦ Zero Trust Network Access (ZTNA)
- ♦ Secure Web Gateway (SWG)
- ♦ Firewall Security (FWaaS)
- ♦ Intrusion Prevention Systems (IPS)
- ♦ Flexibility, reduced complexity, increased performance

Secure Access Service Edge (SASE) – defined by [Gartner](#) – is a security framework prescribing the integration of security and network connectivity technologies into a single cloud-delivered platform to enable secure and fast cloud transformation

Summary

- Dynamic Routing based on Link quality measurements
- Link Load balancing
- Increased bandwidth
- Enhanced Security
- Cloud ready access – SASE
- Application visibility
- Scalability
- Cost optimization
- Improved high availability

Thank You

Q&A